

$$R.E = (b^*ab^* + a^*ba^*)$$

$$\begin{array}{cc} \downarrow & \downarrow \\ \lambda a \lambda & \lambda b \lambda \\ \geq a & b. \end{array}$$

⑨ All words that don't end at b.

$$(a+b)^* a$$

17-10-2024

Write a regular expression for
→ All words that start and end with
different letters over

$$\Sigma = \{x, y\}$$

$$\{xy, yx, xxxxy, xyxxx, yxyx, yxyxyx, \dots\}$$

$$R.E = x(x+y)^*y + y(x+y)^*x$$

Give a regular expression

$$\Sigma = \{0, 1\}$$

1) W/W begins with 1 and end with a zero.

$$\{10, 1000, 11110, 0011101110, \dots\}$$

$$R.E = 1(0+1)^*0$$

→ such that

2) W/W contains exactly three 1's

$$\{0000, 111, 0111, 01110, 111000, \dots\}$$

~~1/11111111~~

~~1/1111011~~

$L = \{ 111, 01011, 10101, 1000110, 110100000, 0000100010001000 \}$

$$R.E = (0^*10^*10^*10^*)$$

3) W/W contain at least three 1's

$L = \{ 111, 0101110, 101010, 11010, 11011, \dots \}$

~~R.E = (0+1)^*111(0+1)^*~~

~~R.E = (0+1)^*1(0+1)^*1(0+1)^*1(0+1)^*~~

$$R.E = (0+1)^*1(0+1)^*1(0+1)^*1(0+1)^*$$

4) W/W begin with one and contain exactly two zero.

~~10011011011011~~

$$1 \ 1^*0 \ 1^*0 \ 1^*$$

5) $W \in \{0, 1\}^*$ | W contains the substring 01

$L = \{ 01, 101, 001, 1001, 100001, 0010100, \dots \}$

$$R.E = (0+1)^*01(0+1)^*$$

~~6) $\{w \in \{0,1\}^* \mid w \text{ contains } 11\}$~~

6) $\{w \text{ in } \{0,1\}^* \mid \text{second symbol of } w \text{ is a } 1\}$

$L = \{1100, 0100, \dots\}$

$R.E = (0+1)1(0+1)^*$

7) $\{w \in \{0,1\}^* \mid |w| < 4\}$

$R.E = (0+1+1) + (0+1)(0+1) + (0+1)(0+1)(0+1)$

8) $\{w \text{ in } \{a,b\}^* \mid |w| \text{ is odd}\}$

$L = \{a, aba, bab, aab, bbb, aaa, \\ aaaabb, baaabb, abaabaabb, \\ \dots\}$

$R.E = (a+b)((a+b)(a+b))^*$

9) $\{w \text{ in } \{a,b\}^* \mid |w| \text{ is even}\}$

$L = \{\lambda, aa, ab, ba, bb, aabb, \\ bbaa, bbbaab, abbbba, \\ \dots\}$

$R.E = ((a+b)(a+b))^* \rightarrow \text{null included}$

$((a+b)(a+b))^* \rightarrow \text{null not included}$

10) $\{w \text{ in } \{a, b\}^* \mid w \text{ does not have } ab \text{ as a substring}\}$
 $= (b^* a^*)$

"Regular Expression" Identities:-

- 1) $u + u = u$
- 2) $u^* = (u^*)^*$
- 3) $u(v + w) = uv + uw$
- ⋮ (slides) . . .

Regular language

language finite
 $\rightarrow$ regular

$\rightarrow$ A language that can be defined by regular expression.

$$L = \{aa, bb\}$$

$$R.E = aa + ab$$

$L_1 =$ All words that start with a

$$\Sigma = \{a, b\}$$

$$R.E = a(a+b)^*$$

$$L_1^* = (a(a+b)^*)^*$$

$$L_2 = b(a+b)^*$$

$\rightarrow$ union

$$L_1 \cup L_2 = (a(a+b)^* + b(a+b)^*)$$

agr koi language regular hai to un ka closure concatenate or union to wo bhi regular ho ga.

→ finite language is regular.

example :-

$$L = \{ aa, bb, ab, ba \}$$

$$R.E = \{ (a+b)(a+b) \}$$

OR

$$= \{ aa + bb + ab + ba \}$$

Equivalent Regular Expression

↳ Two regular expressions are equivalent if they generate same language!

→ sort of graph.

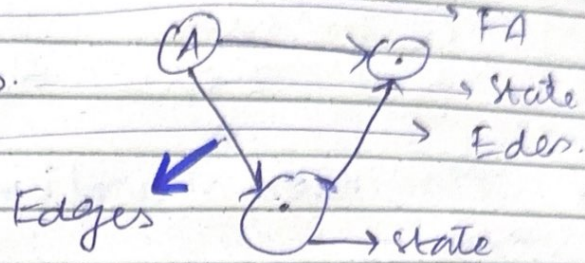
22-10-2024

Finite Automata (FA) → graph.

→ 10 states

Graphs
Nodes/Vertices.

Edges
transitions.



- 1) States: → Represented in the form of circle.
- 2) Exact one start state.
- 3) One or more final states (May be None)

* $\ominus$] → initial/Beginning state.

* $\oplus$] → final state.

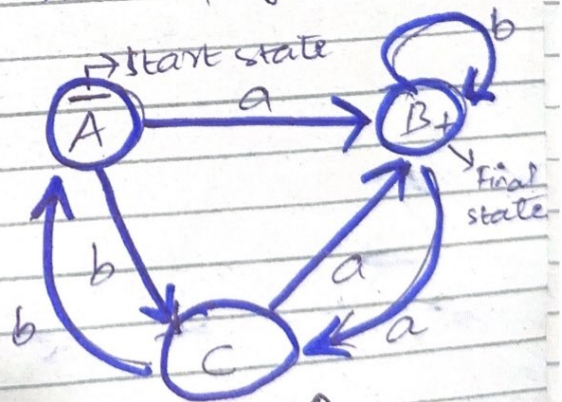
→ used at Edges

- 4) Σ (set of Alphabet)
- 5) Transition function.

(Kis state se kia read krna
na or kis state p jana hai)

$(A, a) \rightarrow B$

→ edges are labelled
with sigma Σ .
e.g. $\Sigma(a, b)$



States → finite

→ Q represent state.

→ F represent final state.

$Q = \{A, B, C\}$
 $F = \{B\}$

$F \subseteq Q$

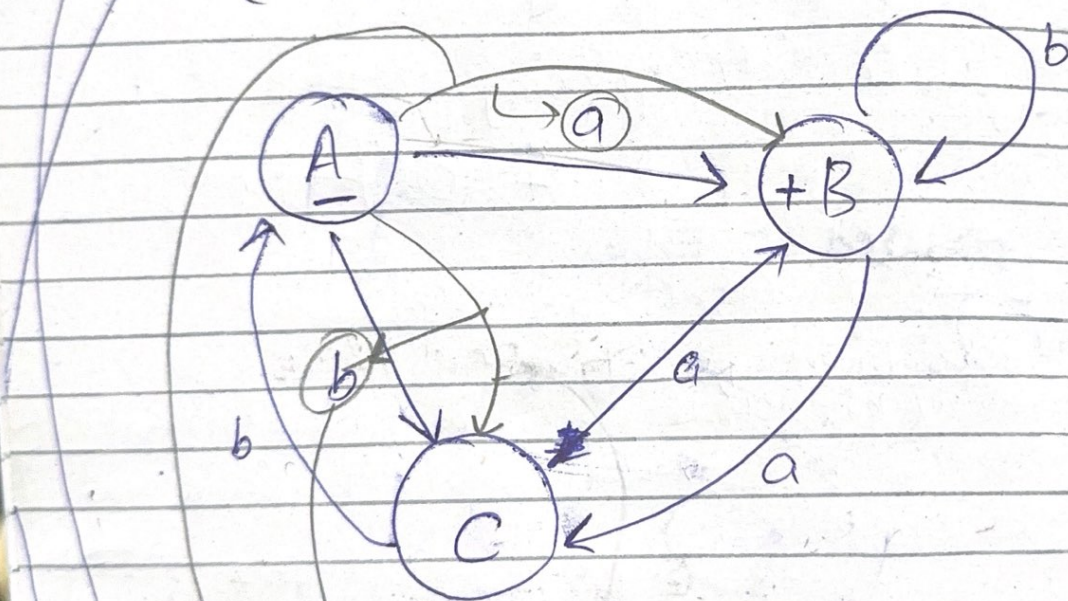
→ Only one character is allowed at each edge. that will be from Sigma Σ .

5. Transition Function.

Start

State works as memory character.

- 1) $(A, a) \rightarrow B$
- 2) $(A, b) \rightarrow C$
- $(B, a) \rightarrow C$
- $(B, b) \rightarrow B$



A pe a read kya to B p chala
gea.

A pe b read kya to C p chala
gea

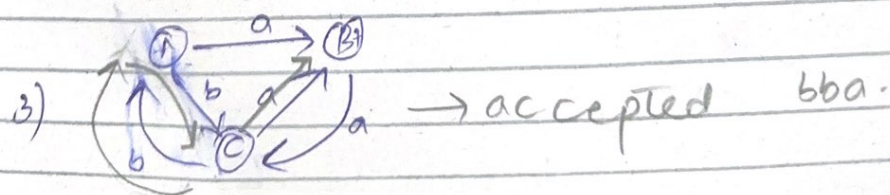
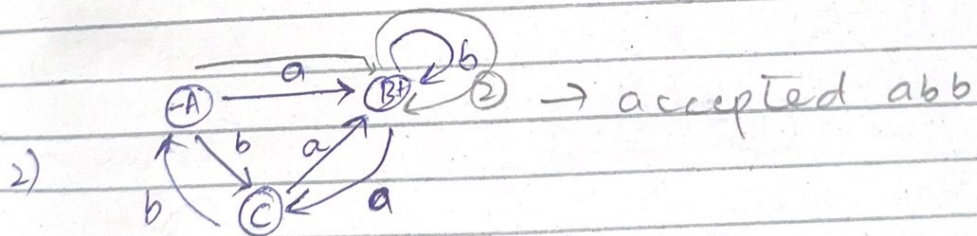
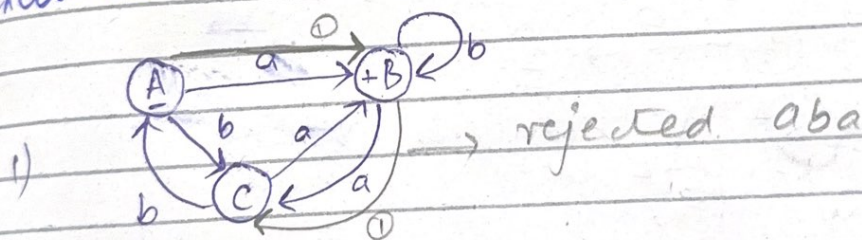
Acceptance and Rejection of Word.

5/Agpt / Control

→ After the execution of string if the control of the string is ended at final state that it will be accepted otherwise it will be rejected.

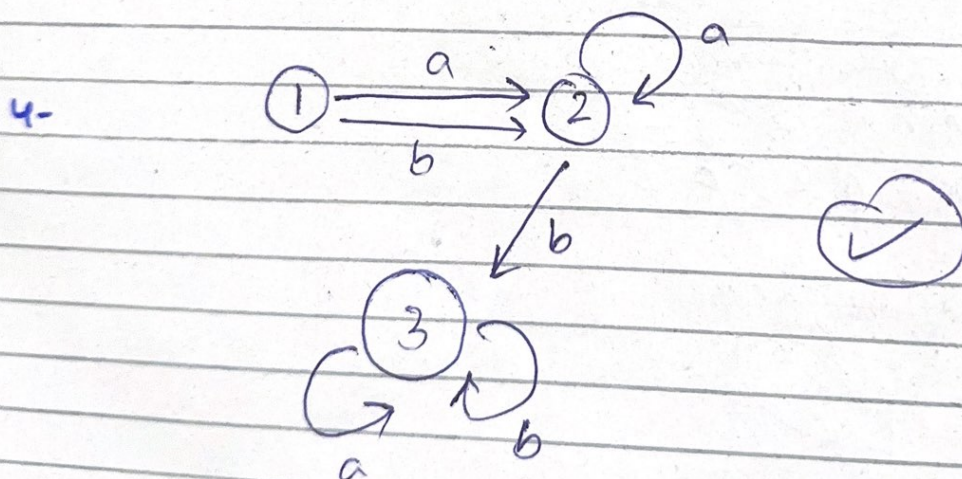
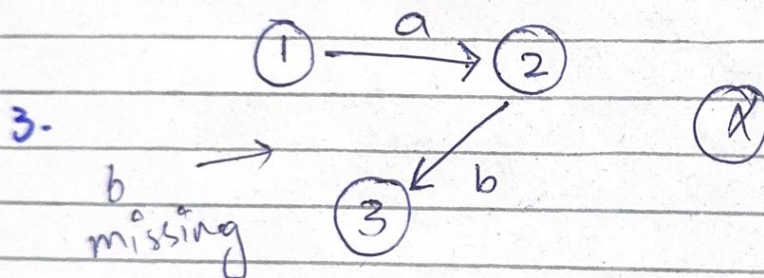
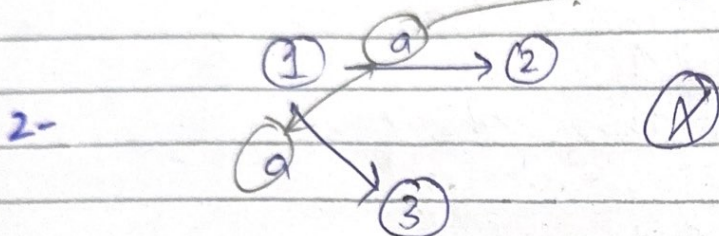
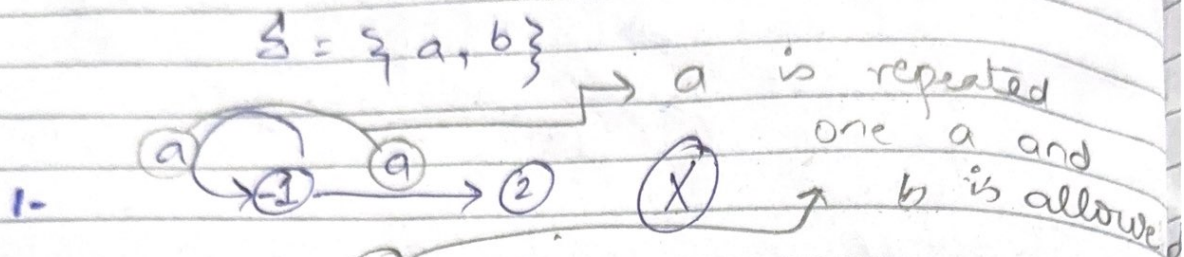
Example: valid (control B p k l u m h o r a h h a).

Random Machine words execution. → { aba, abb, bba }
 rejected control C p end h o r a h h a. ↓ final state.
 valid.



→ No duplicate Edge and no missing edge.

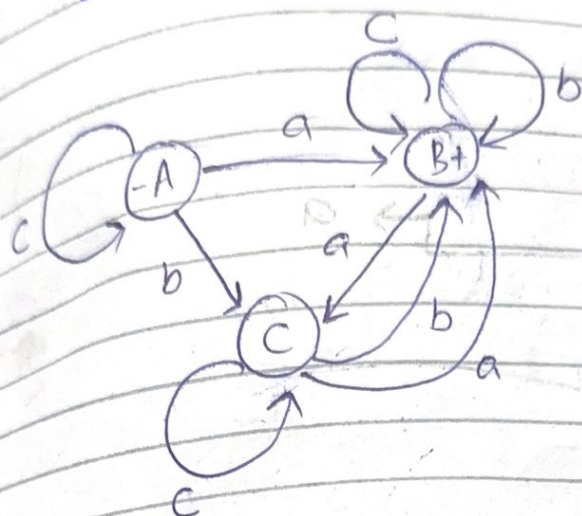
$$S = \{a, b\}$$



$\Sigma = \{a, b, c\}$

$\rightarrow$ agr 3 han to
hr state

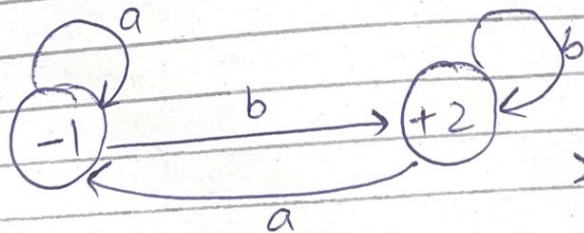
se 3 pri
change
pr repeat
na
hon.



Example: if $\Sigma = \{a, b\}$

$L =$ All words that end at b.

$F = \{ b, ab, bb, bbb, aab, abb, abab, bab, abbb, \dots \}$

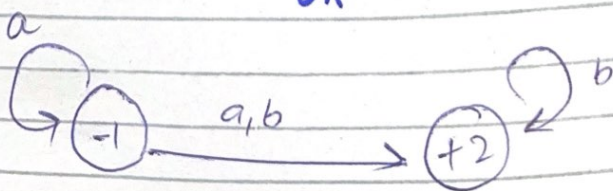


* No invalid word is accepted.

* No duplicate edge and missing edge.

* All valid words are accepted.

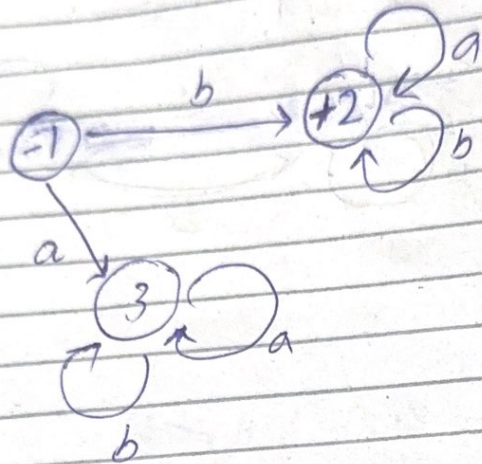
OR



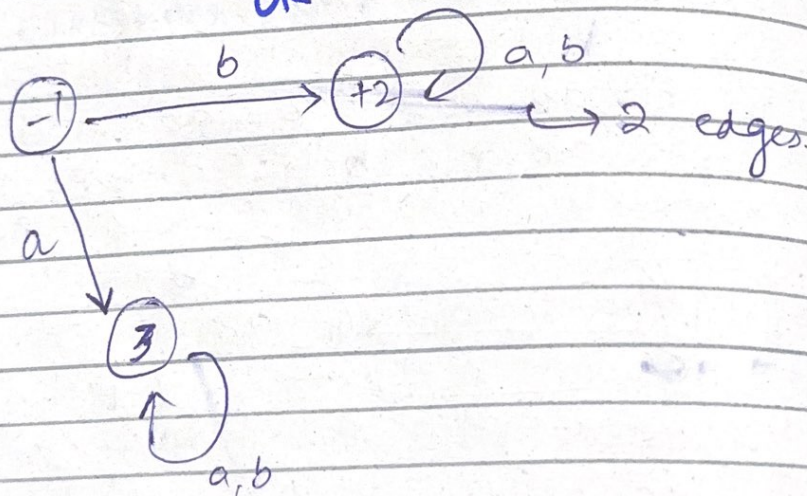
$\Sigma = \{a, b\}$

All words that start with b.

$F = \{b, bb, bbb, bba, bab, babb, baaab, \dots\}$



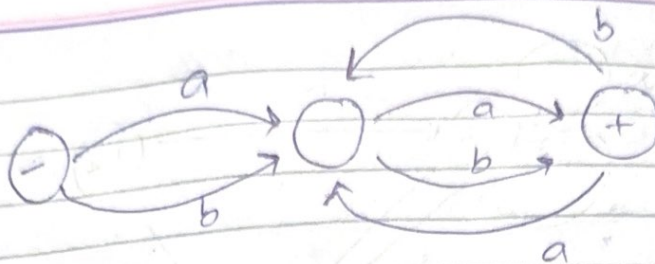
OR



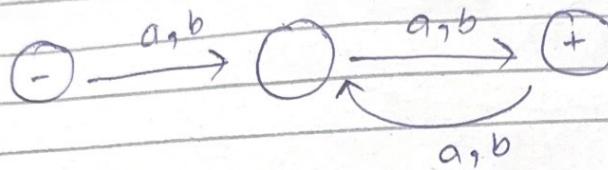
All words that have even length.

$\Sigma = \{a, b\}$

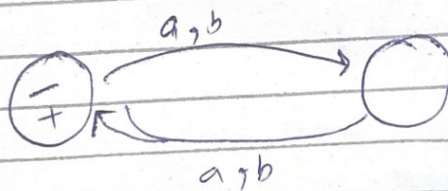
$F = \{aa, bb, ab, ba, aaaa, bbbb, bbaa, baabb, \dots\}$



OR



OR



(null will also be accepted).

24-10-2024

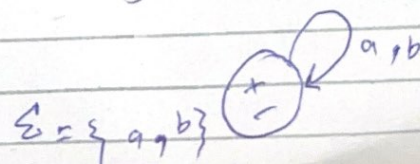
DFA

(-) initial state → ○

(+) final state ⊙

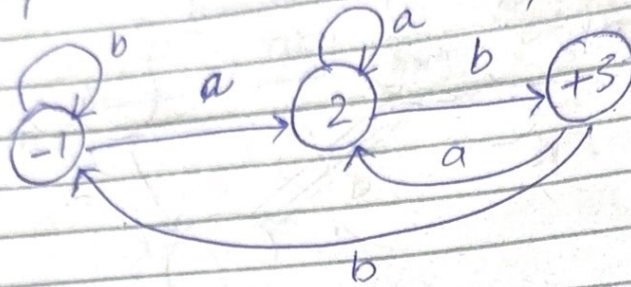
○ intermediate state

→ Each FA contain exactly one start state.



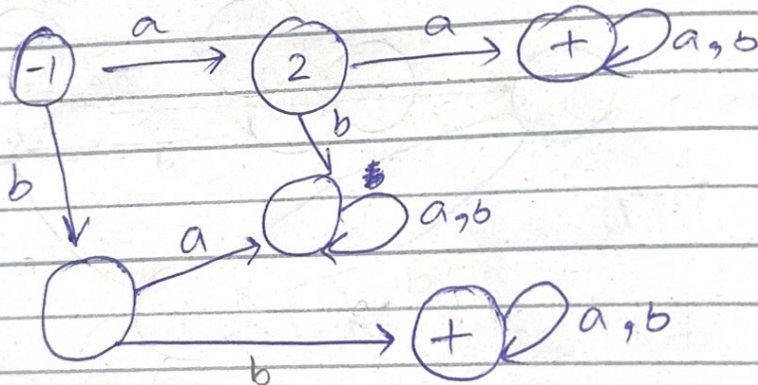
All words that ends at ~~ab~~ ab.
 $\Sigma = \{a, b\}$

$L = \{ab, aab, bab, aaab, baab, bbab, \dots\}$



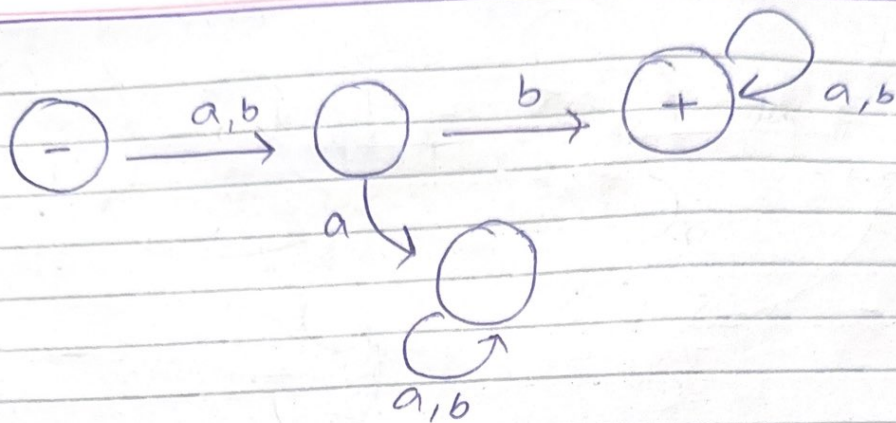
All words that start with double letter.
 $\Sigma = \{a, b\}$

$L = \{aa, bb, aab, bba, aaa, bbb, aabb, aaba, bbab, aaabbbbaa, \dots\}$



All words whose second letter is b
 $\Sigma = \{a, b\}$

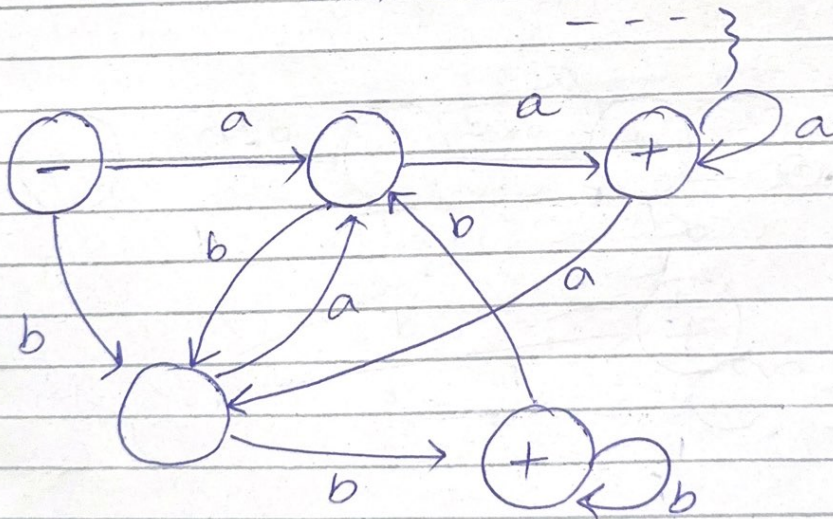
$L = \{ab, bb, aba, bba, bbb, abb, bbaa, abaa, \dots\}$



All words that end at double letter.

$$\Sigma_1 = \{a, b\}$$

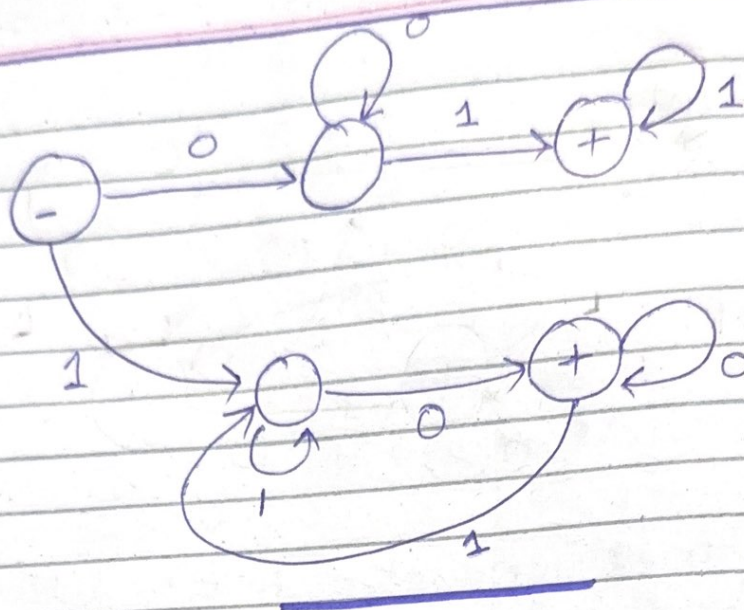
$$L = \{aa, bb, aaa, bbb, abb, baa, aabb, bbba, babb, abaa, \dots\}$$



All words that start and end with different letter.

$$\Sigma_2 = \{0, 1\}$$

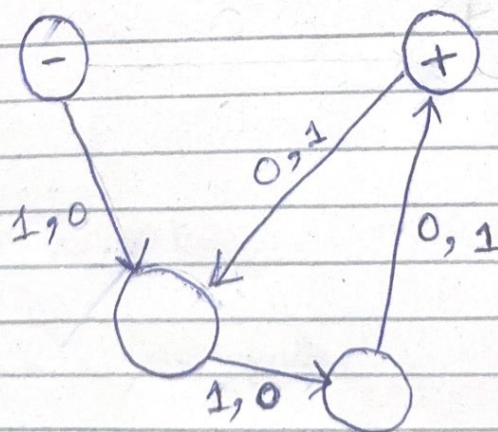
$$L = \{01, 10, 001, 100, 0011, 1100, 0101, 1010, \dots\}$$



29-10-2024

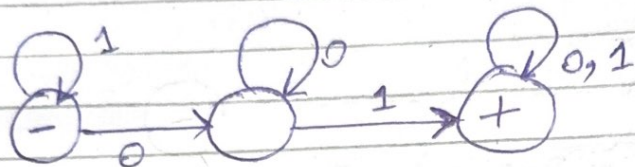
All words whose length is multiple of 3.
 $\Sigma = \{0, 1\}$

$L = \{ 000, 111, 010, 101, 001, 100, 000000, 111111, 100110, 0110010, 110011, \dots \}$



All words that contain 01 as a substring. $\Sigma = \{0, 1\}$

$L = \{ 01, 001, 101, 1001, 0001, 100001, 101001, 100100001, \dots \}$



Formal Def of F.A:

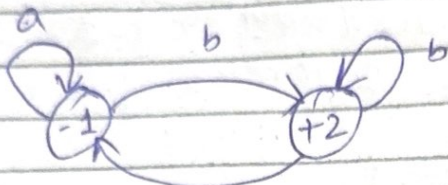
$$FA = (Q, \Sigma, q_0, F, \delta)$$

- 1) $Q = \{q_0, q_1, q_2, \dots, q_n\}$ where n is finite.
- 2) Σ = set of input alphabets.
- 3) q_0 = start state.
- 4) δ = transition function.

$$\delta(q_i, x_j) = q_k$$

- 5) $F \subseteq Q$ is the set of final states F may be $\emptyset$.
- 6) $F \rightarrow$ set of final state.

Transition Table.

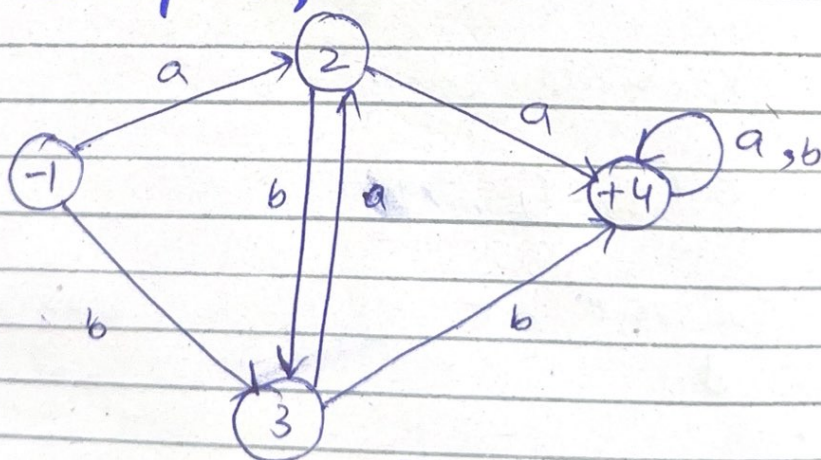


	a	b
-1	1	2
+2	1	2

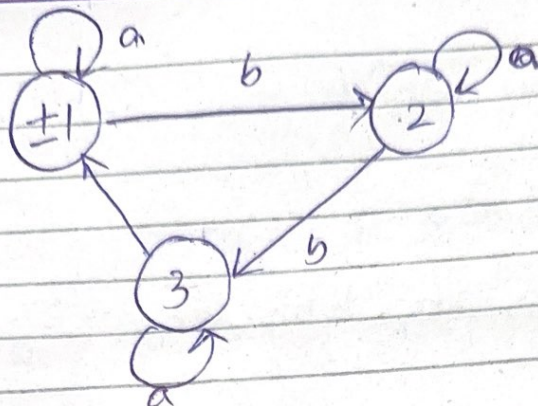
* -1 state pr a ja rah. ha 1 state pr b ja raha ha 2 pr state p.

All words that contain double letter as substring.

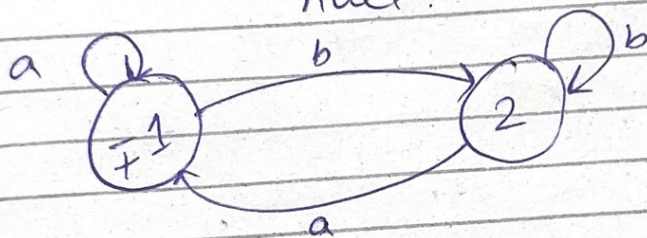
$$\Sigma = \{a, b\}$$



All words that contain b, ~~but~~ then b should be multiple of 3.

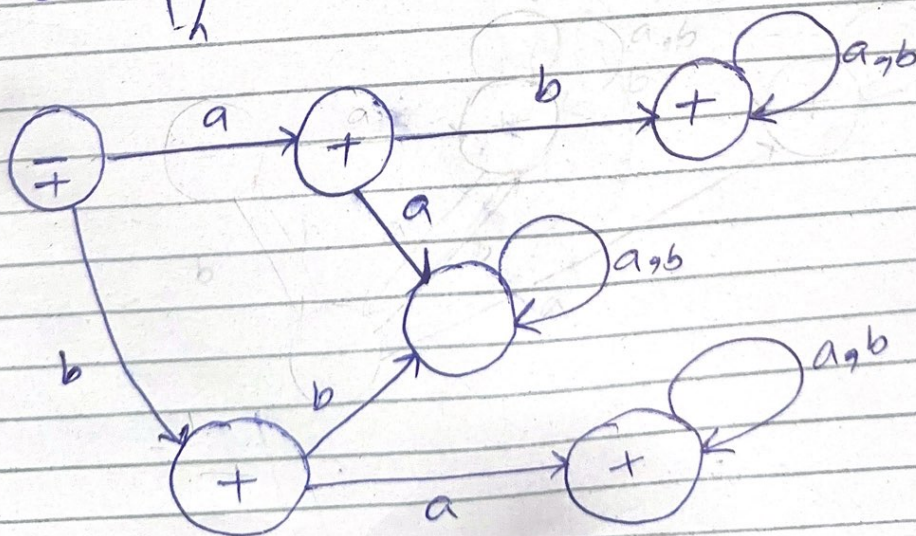


All words that end at a. including null.



All words that donot start with double letter.

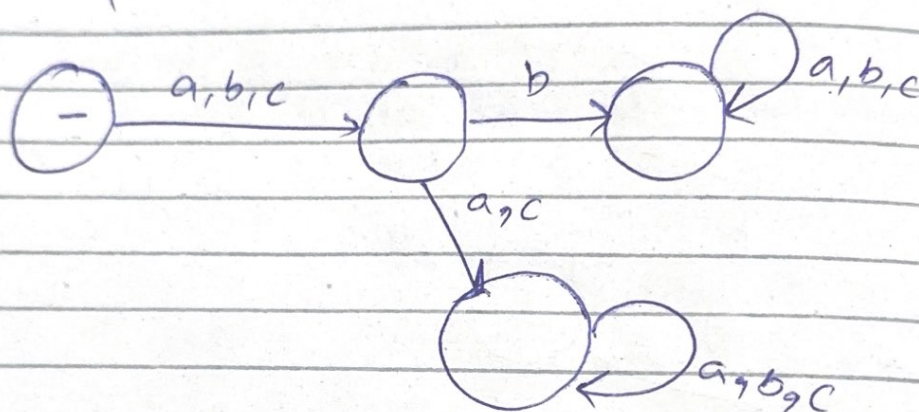
$\lambda, a, b,$
 $L = \{ ab, ba, aba, baa, bab, abb, \dots \}$



All words in which the second letter is b.

$$\Sigma = \{a, b, c\}$$

$$L = \{ab, cb, bb, abc, bbc, cba, \dots\}$$



All words having even number of a's and even number of b's

